

2(a)	+ q : thermal energy transfer to system	B1																
	+ w : work done on system	B1																
2(b)(i)	($W =$) $2.6 \times 10^5 \times (3.8 - 2.3) \times 10^{-3} = 390 \text{ J}$	A1																
2(b)(ii)	no (total) change (in internal energy)	B1																
	gas returns to its original temperature	B1																
2(c)	A to B row all correct (1370, – 390, 980)	B1																
	B to C row all correct (0, 550, 550)	B1																
	C to A row: ΔU adds to the other two ΔU values to give zero	B1																
	C to A row: $w = 0$ and q adds to w to give ΔU value	B1																
	complete correct answer:																	
	<table><tr><td>change</td><td>q / J</td><td>w / J</td><td>$\Delta U / \text{J}$</td></tr><tr><td>A to B</td><td>(+)1370</td><td>–390</td><td>(+)980</td></tr><tr><td>B to C</td><td>0</td><td>(+)550</td><td>(+)550</td></tr><tr><td>C to A</td><td>–1530</td><td>0</td><td>–1530</td></tr></table>	change	q / J	w / J	$\Delta U / \text{J}$	A to B	(+)1370	–390	(+)980	B to C	0	(+)550	(+)550	C to A	–1530	0	–1530	
change	q / J	w / J	$\Delta U / \text{J}$															
A to B	(+)1370	–390	(+)980															
B to C	0	(+)550	(+)550															
C to A	–1530	0	–1530															